Hospital Readmission Analytics and Prediction Using Public Data

Data Engineering and MLOps, Individual Project (Project 5), Parts 1 and 2
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> **Academic decision-support prototype.** The model described here is not clinically validated and must not be used for real patient decisions. Only public, de-identified data and made-up example inputs were used.

1. Objective and problem understanding

Hospital readmissions within 30 days of discharge are costly and are often used as a quality indicator. The project builds a privacy-conscious data pipeline over a public, de-identified hospital dataset and then extends it into an MLOps workflow that estimates readmission risk. Part 1 delivers a reproducible ETL pipeline, a PostgreSQL analytical mart, Airflow orchestration and a Streamlit dashboard of admission and readmission patterns. Part 2 adds a readmission-risk model (comparison of Logistic Regression, Random Forest and XGBoost), MLflow tracking and registry, a FastAPI inference service, a Docker image, dashboard integration, monitoring and a documented model lifecycle.

The prediction task is defined by the dataset: the positive class is an encounter followed by readmission within 30 days (`readmitted == "<30"`); readmission after 30 days (`">30"`) and no readmission (`"NO"`) are negative. The intended prediction time is at discharge, so only information available by then is used.

2. Dataset source and access

Item	Details
Dataset	Diabetes 130-US Hospitals for Years 1999–2008 (Strack et al., 2014), UCI Machine Learning Repository, dataset 296, licence CC BY 4.0
Access	https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008 — free download, no account. Copy `diabetic_data.csv` and `IDS_mapping.csv` into `data/raw/`. The unmodified files are also in the repository, so a fresh clone needs no download.
Content	101,766 inpatient encounters of diabetic patients, 50 columns: demographics (age band, gender, race), admission type/source, discharge disposition, length of stay, procedures, medications, up to three ICD-9 diagnoses, lab results (HbA1c, glucose) and the readmission label
Privacy	De-identified at source. The pipeline replaces the patient number with a SHA-256 surrogate key (`patient_sk`); no names, dates or addresses exist. Race and payer are never model inputs; race is used only for aggregate fairness checks.

3. Architecture

The architecture follows the layers suggested in the brief: source, ingestion, raw/staging, transformation, storage, analytics and an MLOps layer. The architecture diagram is `docs/architecture.png` (Figure 1) and the mart schema is Figure 2. Figure 3 shows how Part 2 attaches to Part 1 without changing it.

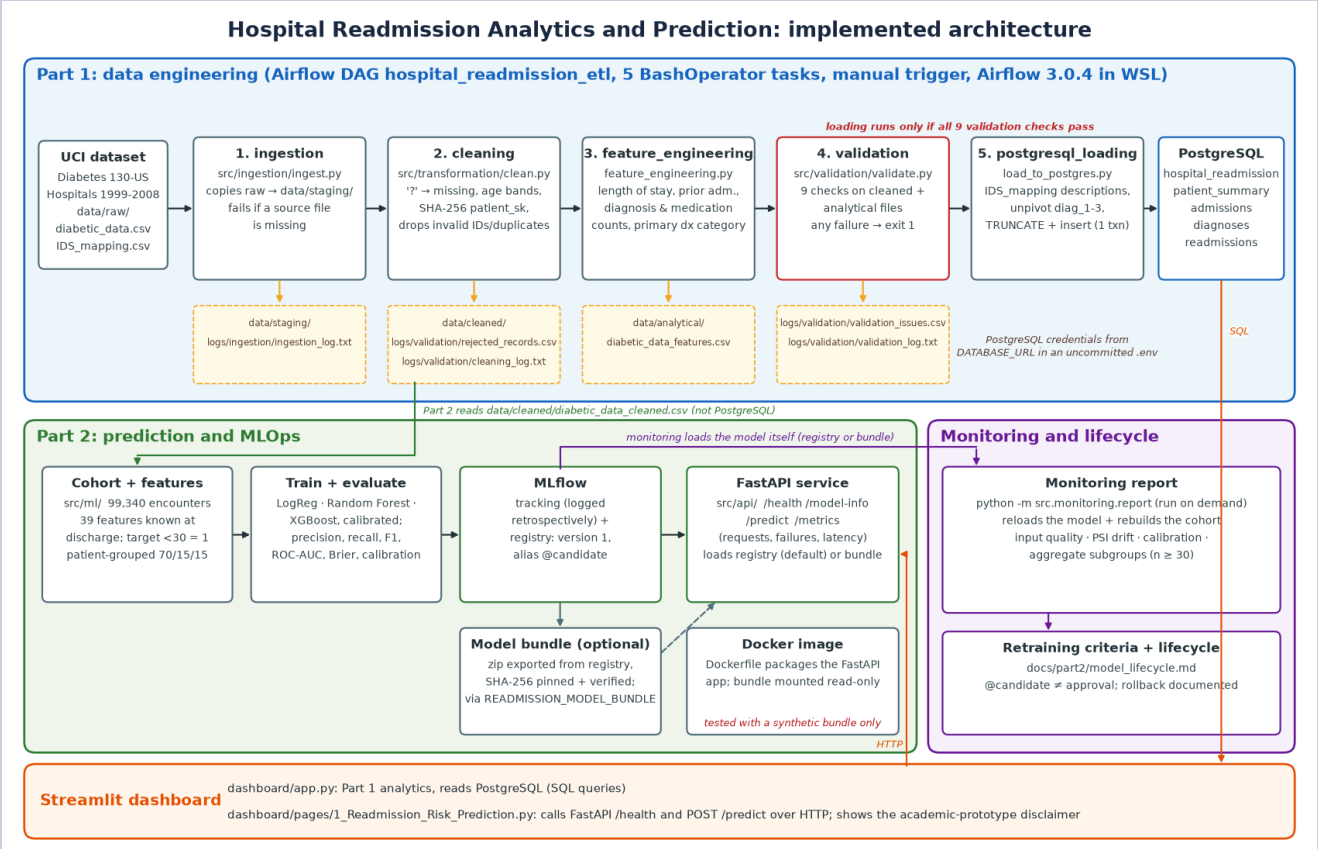


Figure 1. Implemented architecture: Airflow tasks and their outputs, PostgreSQL, Part 2 MLOps and the dashboard (`docs/architecture.png`).

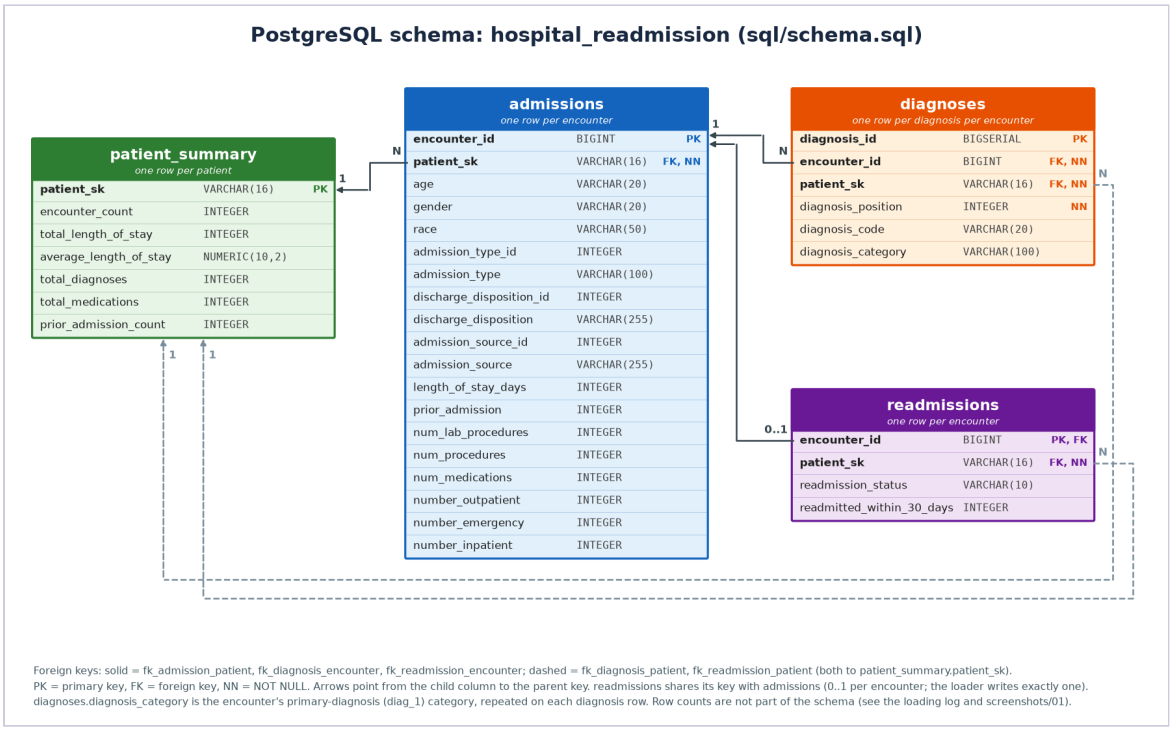


Figure 2. PostgreSQL schema of the analytical mart, drawn from `sql/schema.sql` (`docs/Schema_Diagram.png`).

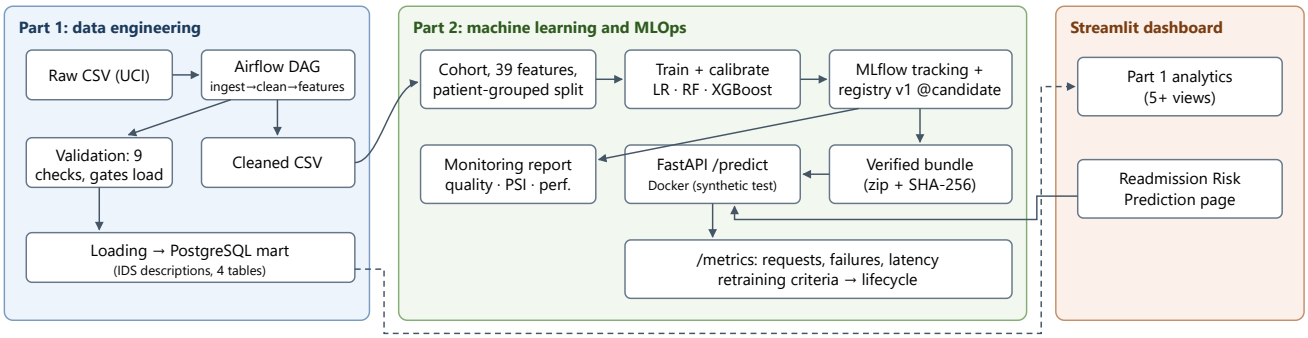


Figure 3. End-to-end architecture. Part 2 reads Part 1's cleaned CSV and leaves the Part 1 pipeline, database and dashboard page unchanged; the prediction page calls the API over HTTP.

4. Part 1: ingestion, ETL, validation and storage

4.1 Ingestion

`src/ingestion/ingest.py` reads the raw CSV files with Pandas, keeps `data/raw/` untouched, copies them to `data/staging/` and writes `logs/ingestion/ingestion_log.txt` with the extraction time, source files, file status and row and column counts (101,766 rows, 50 columns). A missing source file stops the run with an explicit error, so Airflow marks the task failed.

4.2 Cleaning, transformation and feature engineering

- **Special missing-value codes:** the dataset's `?` code is replaced by a true missing value; categorical blanks are standardised.
- **Standardisation:** age bands (e.g. `[60-70)` → `60-70`), diagnosis codes grouped into categories, medication fields normalised, and the readmission label kept as `<30`, `>30`, `NO` with a 0/1 30-day indicator.
- **Invalid and duplicate encounters:** records missing required identifiers are rejected and duplicate `encounter_id`s removed; both are written to `logs/validation/rejected_records.csv` (header only when nothing is rejected).
- **Patient-safe surrogate key:** `patient_sk` = first 16 hex characters of SHA-256 of the patient number (71,518 unique keys, 0 duplicates).
- **Engineered variables:** length of stay, prior admissions, diagnosis count, medication count and primary diagnosis category (`src/transformation/feature_engineering.py`).

4.3 Validation and error log

`src/validation/validate.py` runs nine checks: required identifiers, duplicate encounters, remaining special codes, valid readmission categories, length-of-stay range, prior-admission flag, diagnosis count, medication count, and matching cleaned and analytical row counts. Each failed check is written as one row (check, message, count) to `logs/validation/validation_issues.csv`, a separate file from cleaning's rejected rows, and the run status to `validation_log.txt`. If any check fails the script exits with code 1, so Airflow marks the validation task failed and the PostgreSQL loading task is not run. On the current data the status is **PASSED** with 0 rejected records. Validation rules are documented in the *Validation Rules* sheet of `docs/data_dictionary.xlsx`.

4.4 PostgreSQL mart and orchestration

`sql/schema.sql` defines the mart in database `hospital_readmission`, loaded by `src/loading/load_to_postgres.py` using `DATABASE_URL` from an uncommitted `.env` file. Loading adds the admission type, discharge disposition and admission source descriptions from `IDS_mapping.csv`, unpivots `diag_1`–`diag_3` into `diagnoses` rows (dropping unknown codes), and reloads all four tables with TRUNCATE and insert in one transaction.

Table	Grain	Rows (read-only query, 15 Oct 2026)
`admissions`	one row per encounter: admission, stay and discharge attributes	101,766

`diagnoses`	one row per encounter diagnosis (up to 3)	303,496
`patient_summary`	one row per surrogate patient: encounter and readmission totals	71,518
`readmissions`	one row per encounter: readmission status and 30-day flag	101,766

The Airflow 3.0.4 DAG `hospital_readmission_etl` (`dags/hospital_readmission_dag.py`, running in WSL) chains ingestion → cleaning → feature engineering → validation → PostgreSQL loading with `BashOperator` tasks; it is triggered manually and can be re-run at any time. Every task uses the default all-success rule, so a failed task stops all later tasks. Airflow's run history records a successful end-to-end run on 6 September 2026 (Figure 4). On 15 October 2026 the DAG was redeployed to the Airflow DAG folder with the project path corrected, and it parsed with no import errors. The validation gate was tested in a separate throwaway Airflow 3.0.4 setup: a failing validation left `postgresql_loading` as `upstream_failed`, and a passing one let it run. The deployed DAG was then run end to end (run `manual__2026-10-15T09:57:34`): all five tasks succeeded in 66 seconds, validation PASSED with a header-only issues file and 0 rejected rows, and the loader wrote 101,766 / 303,496 / 71,518 / 101,766 rows. For that run the database connection was confined to a separate test schema, so the production tables were not modified, and the regenerated data files were byte-identical to the committed ones. A read-only query on the same day returned the row counts above and confirmed that the live tables, keys and constraints match `sql/schema.sql`. The 30-day readmission rate in the mart is 11.16% (11,357 encounters).

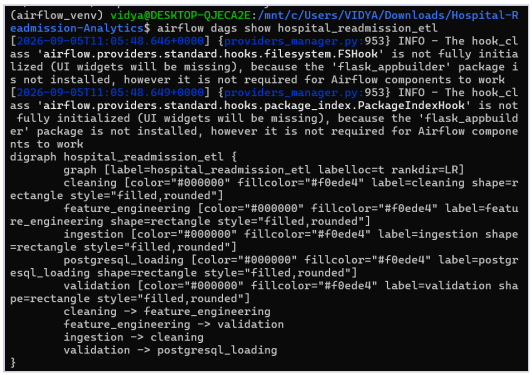


Figure 4. Airflow DAG structure (Part 1 evidence).

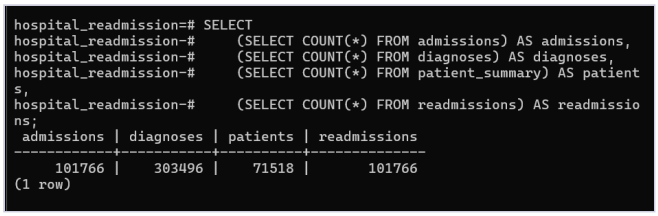


Figure 5. PostgreSQL table row counts (Part 1 evidence).

5. Part 1: dashboard and interpretation

The Streamlit dashboard (`dashboard/app.py`) reads directly from PostgreSQL. It shows three headline indicators (101,766 encounters, 11.16% 30-day readmission rate, 4.40 days average stay) and six charts, each with its underlying data: readmission rate by age group, by diagnosis category and by prior admission history, average length of stay by readmission status, admission type distribution, and the top 10 discharge dispositions. The prior-admission view identifies the main high-risk cohort: in the cleaned data the 30-day readmission rate rises from 8.4% for patients with no inpatient visit in the previous year to 12.9%, 17.4% and 25.7% for one, two and three or more visits. Prior inpatient visits later turn out to be the Part 2 model's most important input. A dedicated high-risk cohort page was not built in Part 1; in Part 2 the prediction page (section 9) fills this role for individual made-up encounters.

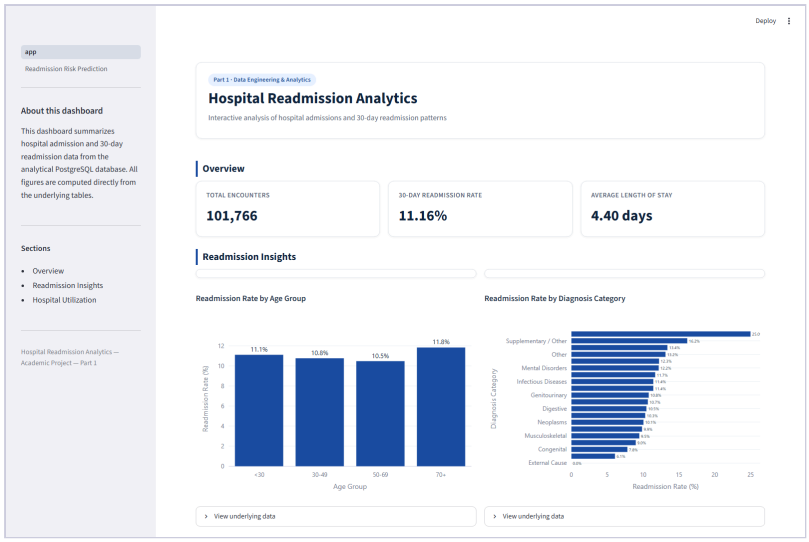


Figure 6. Part 1 dashboard overview, captured on 15 October 2026 from the running app reading PostgreSQL (full page: `screenshots/08_streamlit_dashboard_full_2026-10-15.png`).

6. Part 2: modelling data, split and leakage control

- **Source:** Part 1's cleaned CSV (101,766 encounters), because the mart does not keep the medication, lab and specialty columns.
- **Cohort (99,340 encounters, 11.39% positive):** encounters ending in death (1,652) or hospice (771) are excluded because readmission risk is undefined or not meaningful for them; 3 invalid genders are removed. The API refuses these discharge codes.
- **Features (39, known at discharge):** 10 numeric (age band midpoint, length of stay, lab and other procedures, medications, prior outpatient/emergency/inpatient visits, number of diagnoses, number of diabetes medications) and 29 categorical (gender, admission/discharge codes, specialty, three ICD-9 diagnosis groups, glucose and HbA1c results, medication change, 17 diabetes medications). Each has a documented availability reason in `src/ml/config.py`.
- **Leakage control:** identifiers, the target, race, payer, `weight` (97% missing) and every `patient_summary` column (which aggregates later encounters) are excluded. Preprocessing (imputation, scaling, one-hot encoding with rare-category grouping) is fitted on training data only, and tests check this.
- **Split:** the dataset has no dates, so a chronological split is impossible. A patient-grouped, stratified, hash-based split gives train 69,538 / validation 14,901 / test 14,901 encounters with **zero patient overlap**; its fingerprint (`45788b05…a941`) is checked before every training and monitoring run.
- **Class imbalance** (7.8 negatives per positive) is handled with class weights on training data only; nothing is resampled.

7. Part 2: model development and evaluation

Each model is a scikit-learn pipeline tuned with patient-grouped 5-fold cross-validation on the training split, then calibrated with sigmoid (Platt) scaling using out-of-fold training predictions, because class weighting inflated raw probabilities (mean 0.43–0.47 against a 0.116 base rate). The decision threshold maximises F1 on validation. A selection rule fixed in advance keeps models within 0.01 of the best validation ROC-AUC and PR-AUC and within 0.002 of the best Brier score, then picks the simplest. The test split was used once, for the selected model only.

Validation (14,901 encounters, 1,722 positives)	Logistic Regression	Random Forest	XGBoost
ROC-AUC	0.6618	0.6673	0.6682
PR-AUC (base rate 0.116)	0.2139	0.2192	0.2271
Brier score / ECE (calibration)	0.0984 / 0.0099	0.0979 / 0.0070	0.0976 / 0.0056
Threshold (max F1)	0.1349	0.1618	0.1532
Precision / recall / F1 at threshold	0.211 / 0.412 / 0.279	0.223 / 0.398 / 0.286	0.223 / 0.411 / 0.289

Selection: Random Forest and XGBoost both met the tolerances; the rule chose the simpler Random Forest (200 trees, depth 16, minimum leaf 10, `balanced_subsample`) although XGBoost is slightly better on most metrics. **Held-out test result** (14,901 encounters, 1,704 positives): ROC-AUC 0.6715, PR-AUC 0.2237, Brier 0.0967, ECE 0.0046; at threshold 0.1618, precision 0.213, recall 0.381, F1 0.273 (TN 10,794, FP 2,403, FN 1,054, TP 650). The model flags about one encounter in five and catches about 38% of real 30-day readmissions: useful for ranking, but modest.

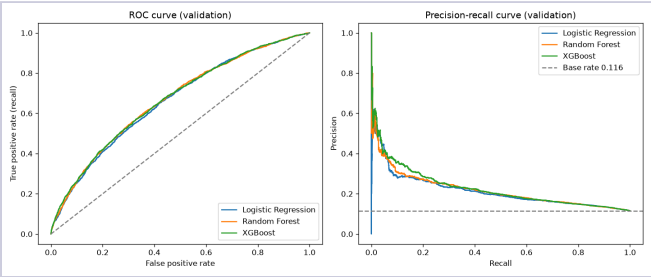


Figure 7. ROC and precision–recall curves on validation.

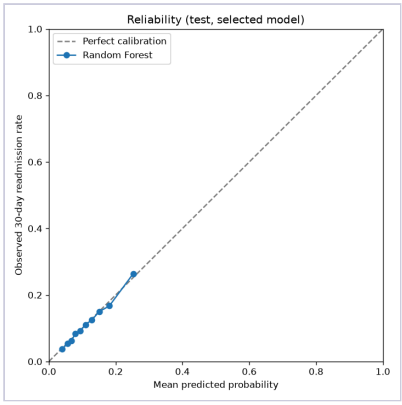


Figure 8. Calibration of the selected model on test.

Fairness (test, threshold 0.1618): ROC-AUC is similar for women and men (0.671 / 0.672), lower for patients aged 70+ (0.635 vs 0.729 under 50), and 0.673 / 0.654 / 0.784 for Caucasian / African American / Hispanic patients; groups with fewer than 30 positives are suppressed. **Explainability:** permutation importance (drop in ROC-AUC when an input is shuffled) ranks prior inpatient visits (0.058) and discharge disposition (0.037) far above all other inputs (each < 0.01). SHAP was not used: it is not installed and, for this calibrated pipeline, would explain encoded columns rather than the 39 inputs. Importance shows association, not cause.

8. Part 2: MLflow tracking, registry and versioning

MLflow 3.17.1 uses a local SQLite store (`mlflow.db`) and artifact folder (`mlartifacts/`), both gitignored. Experiment `hospital-readmission-30d` holds a comparison parent run with child runs for each model's validation results and the selected model's test results (parameters, CV and validation metrics including Brier and ECE, thresholds, confusion matrices, plots and tables), plus a registration run. The saved Random Forest is registered as `hospital-readmission-30d-classifier` **version 1**, stored in the pickle-free skops format, with tags for threshold, model SHA-256, split fingerprint and provenance, and the alias `@candidate`.

> **Disclosure:** the MLflow runs were logged *after* training from the saved Phase 3 result files (tagged `historical_phase3_results`, `retrained=false`); the training script itself does not call MLflow, and individual tuning trials are not logged as runs. The history was deliberately not regenerated.

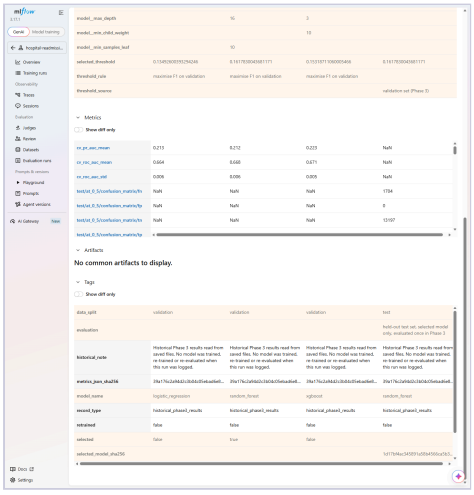


Figure 9. MLflow comparison of the model runs (metrics and provenance tags).

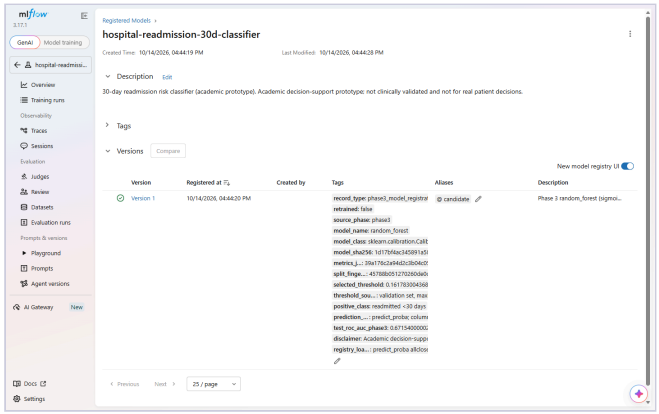


Figure 10. Registry: version 1 with alias `@candidate`.

9. Part 2: deployment – FastAPI, Docker and dashboard integration

API (`src/api/`): `GET /health`, `GET /model-info`, `POST /predict` and `GET /metrics`. At start-up the API resolves `@candidate` and refuses to serve (HTTP 503) unless the version is READY, skops-serialised, has the exact 39-input signature, the expected threshold and the pinned model hashes. Requests are validated strictly (types, ranges, allowed categories; extra fields such as `race` or identifiers are rejected with HTTP 422 without echoing values). Responses return the probability, the high-risk flag, the threshold, the model version and the disclaimer. Request bodies are never logged.

Docker: the `Dockerfile` (Python 3.11 slim, non-root user, health check) contains code and dependencies only; the model is mounted read-only as a bundle exported from the registry. Because the MLflow store holds absolute local paths, the registered version is exported once to `model_bundle/` with a manifest of file hashes, and can be packed into a reproducible zip (`model_bundle_v1.zip`, SHA-256 `5993ef3a…1b04`) so that a fresh clone can serve the exact registered model (`docs/part2/model_bundle.md`). A fresh clone without any MLflow store served the unzipped bundle and passed the smoke test (health 200, model-info 200, predict 200, extra field 422).

Dashboard integration: a separate page, `dashboard/pages/1_Readmission_Risk_Prediction.py`, collects the 39 inputs (no race or payer; expired and hospice codes not offered), calls the API and shows the probability, risk group and threshold with the disclaimer. Part 1's page is unchanged. With made-up inputs, Scenario A (form defaults) gives 9.2%, "Lower risk"; Scenario B (4 prior inpatient visits, discharged to a skilled nursing facility) gives 31.6%, "Higher risk"; with the API stopped the page shows a clear message instead of failing.

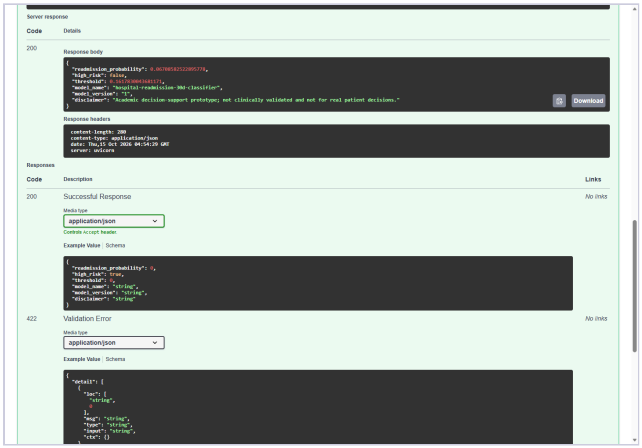


Figure 11. `POST /predict` response (HTTP 200).

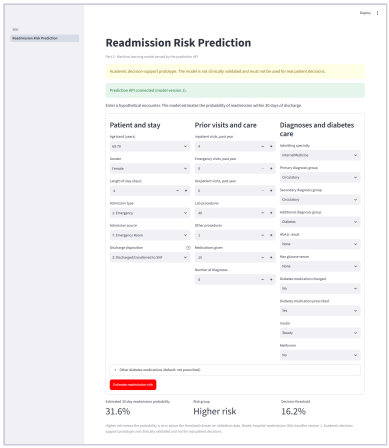


Figure 12. Dashboard prediction page, Scenario B.

10. Part 2: monitoring

Aspect	Implementation (`src/monitoring/`, `src/api/telemetry.py`)	Result on the held-out test batch
Input-data quality	API rejects each invalid request (422); the report counts missing and invalid values per input per batch with the API's own rules; a batch fails above 1% in any input	0 missing, 0 invalid: passed
Data drift	Population Stability Index per input against the training split (10 quantile bins; unseen categories counted); 0.10 moderate, 0.25 significant	No input above 0.25
Prediction drift	PSI of predicted probabilities against validation	0.0021 (stable)
Performance and calibration	ROC-AUC, PR-AUC, Brier, ECE, precision/recall/F1 and flagged share when labels are available	ROC-AUC 0.6715, ECE 0.0046
Subgroups	Metrics by gender, age group and race, aggregate only, groups under 30 suppressed	Lower ranking for age 70+
Latency and failures	`/metrics`: request counts by status, failure rate, p50/p95 latency, prediction distribution, no input values	Shown live in the demo

Real and simulated batches are judged separately. Two clearly labelled *simulated* batches (test rows with age +10 years, +2 prior inpatient visits, +2 days stay) show that the checks react: prediction PSI above 2.0, ECE 0.057–0.079 and drift on the three shifted inputs. **Implemented vs documented:** the checks, report and `/metrics` are implemented and tested; they run on demand. There is no continuously running production monitoring, alerting or persistent metric store (telemetry is in memory and resets on restart).

11. Retraining criteria and model lifecycle

Retraining is recommended (for a person to act on; nothing retrains automatically) when: PSI > 0.25 on one of the 10 most important inputs in two consecutive batches; prediction PSI > 0.25; ROC-AUC more than 0.05 below the validation reference; ECE above 0.03; or the flagged share changes by more than 0.10. A subgroup ROC-AUC more than 0.10 below the overall triggers a fairness review, and a failed input-quality check sends the batch back to the data pipeline first. New labelled data or a change in the Part 1 data are also reasons to retrain. On the real test batch no criterion is met; on the simulated batches retraining is recommended, as intended.

The lifecycle (`docs/part2/model_lifecycle.md`): prepare data → train and select → log and register a new version with `@candidate` → **approval** by a named reviewer (recorded as version tags and an `@approved` alias) → release (update the pinned hashes in a reviewed change, export and archive the bundle) → serve → monitor → retrain as a new version. Versions are never overwritten, so **rollback** means moving the alias back, restoring the previous pinned hashes and redeploying that version's archived bundle. **Important distinction:** `@candidate` means "selected and verified", not "approved". Version 1 has not been approved for any use, consistent with its status as an academic prototype.

12. Testing and reproducibility

The repository contains automated tests for target construction, cohort rules, leakage checks, the patient-grouped split and its fingerprint, preprocessing, training and selection, metrics, MLflow logging and registry, API validation and failure handling, bundle integrity, the Docker configuration, telemetry, monitoring and input quality, and the dashboard page. On 15 October 2026 the full suite was run on the final code (with the real MLflow store present): **260 passed, 0 failed, 0 skipped** in 6 min 45 s. A previously flaky telemetry privacy test, which searched the raw metrics text and could match the digits of a measured latency, was rewritten to compare exact keys and values, and two stricter tests were added; the telemetry tests then passed 25 of 25 repeated runs. Part 1 was also re-run end to end (ingestion, cleaning, feature engineering, validation) in a fresh clone: the staged, cleaned and analytical files were identical in content to the committed ones and validation PASSED. All seeds are 42; the full XGBoost run differed by about 1e-3 between two machines, so results are reproducible per environment rather than bit-for-bit across platforms.

13. Limitations

- **Modest discrimination** (test ROC-AUC 0.67, PR-AUC 0.22): most readmissions are missed at any threshold that keeps false alarms manageable; the F1 threshold is a policy choice.
- **No time-based validation:** the data has no dates, so performance on future patients is not measured; data are from US hospitals in 1999–2008.
- **Real model not tested inside Docker:** Docker is not installed on the project machine. The image was built and tested with a synthetic bundle in a separate environment (base image pulled through a registry mirror); the real bundle was served with Uvicorn only.
- **Retrospective MLflow logging** and no formal approval of version 1 (see sections 8 and 11).
- No SHAP or per-patient explanations; permutation importance is global.
- The API has no authentication, HTTPS or rate limiting, and telemetry is in memory: a local prototype only.
- The Airflow DAG's project path is fixed in the DAG file and must be edited if the project folder moves.

14. Conclusion

The project delivers a reproducible, documented pipeline from a public de-identified dataset to an analytical PostgreSQL mart and dashboard, and extends it into an end-to-end MLOps prototype: leakage-aware features, a patient-grouped split, a fair comparison of three calibrated models, MLflow tracking and a versioned registry

entry, a verified FastAPI service with a Docker image, dashboard integration, monitoring of input quality, drift, calibration, subgroups, latency and failures, and explicit retraining and lifecycle rules. The model's predictive performance is modest and it is suitable only as an academic decision-support demonstration, not for clinical use.

Reference. Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: analysis of 70,000 clinical database patient records. *BioMed Research International*, 2014, 781670. Dataset: UCI Machine Learning Repository, Diabetes 130-US Hospitals for Years 1999–2008, doi:10.24432/C5230J.